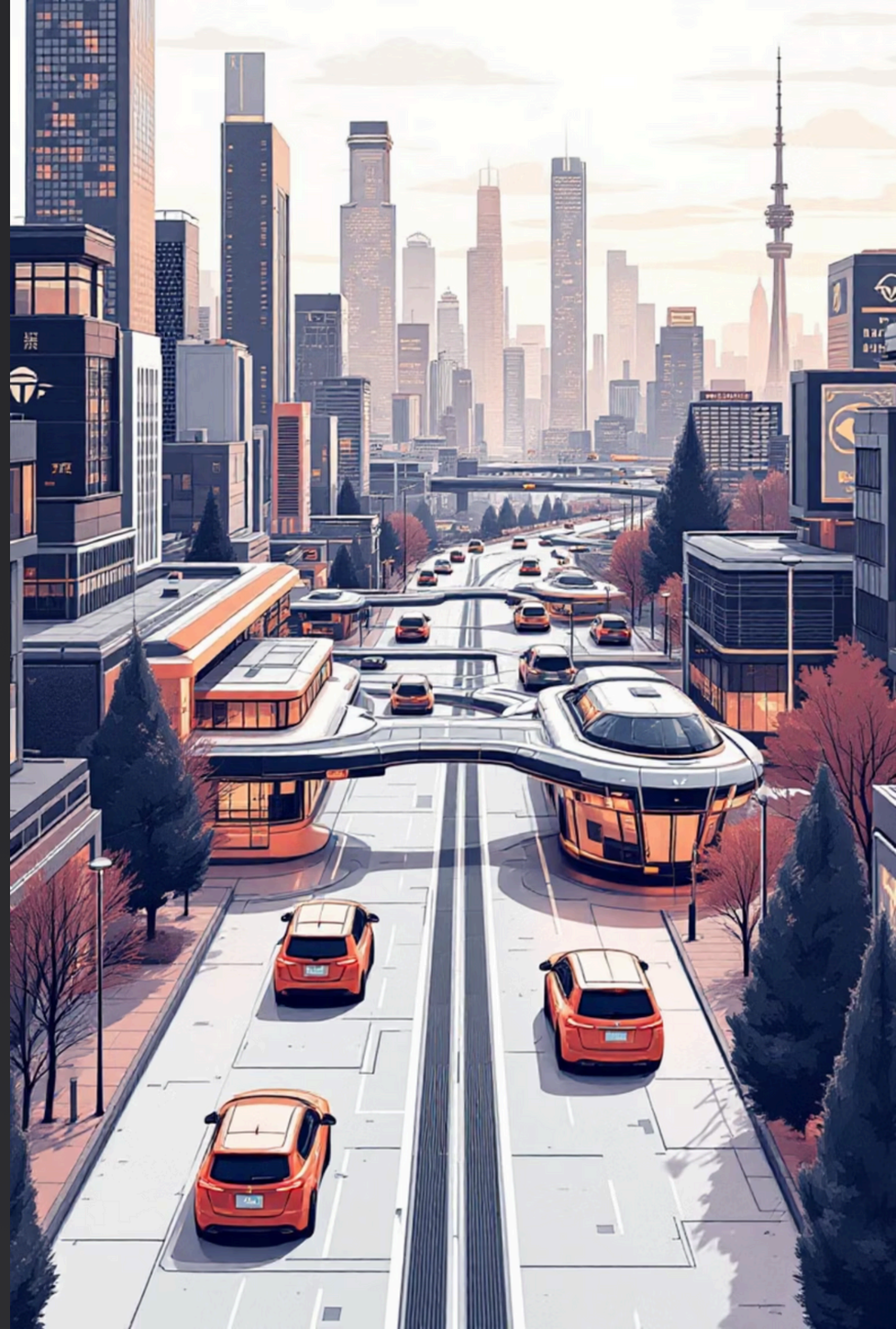
Optislot (Just Automated Real-time Intelligent Vehicle Parking System)

Smart Cities & Intelligent Transportation Systems

By Reshekumar V

Department of Computer Science and Engineering, KPR Institute of Engineering and Technology



Problem: The Parking Predicament

Inefficiency

High search time for parking causes congestion and fuel waste.

High CAPEX

Traditional systems need physical sensors at *every* slot (1:1 ratio).

Maintenance

Decentralized hardware increases failure points.

Location: Urban centers, malls, and corporate campuses.

Importance: Critical to reduce infrastructure costs, traffic and carbon emission.

Existing Scenario: The Status Quo

Current Methods:

- Slot-Sensors: Ultrasonic/magnetic loops at the Entrance Gate.
- Manual: Security guards directing traffic.

Drawbacks:

- Cost: Installing 100s of sensors is expensive.
- Rigidity: Hard to scale; requires physical cabling for every new spot.

Insufficiency: Relies on "Brute Force Hardware" instead of scalable software logic.



Proposed Solution: Optislot's Innovation

Core Idea: Centralized **Sensor Fusion** at the entry gate, removing slot-level sensors.

01

Optical Profiling

Camera extracts the Registration Number from the Number plate.

02

Logic

"Bin-Packing Algorithm" allocates the optimal slot.

03

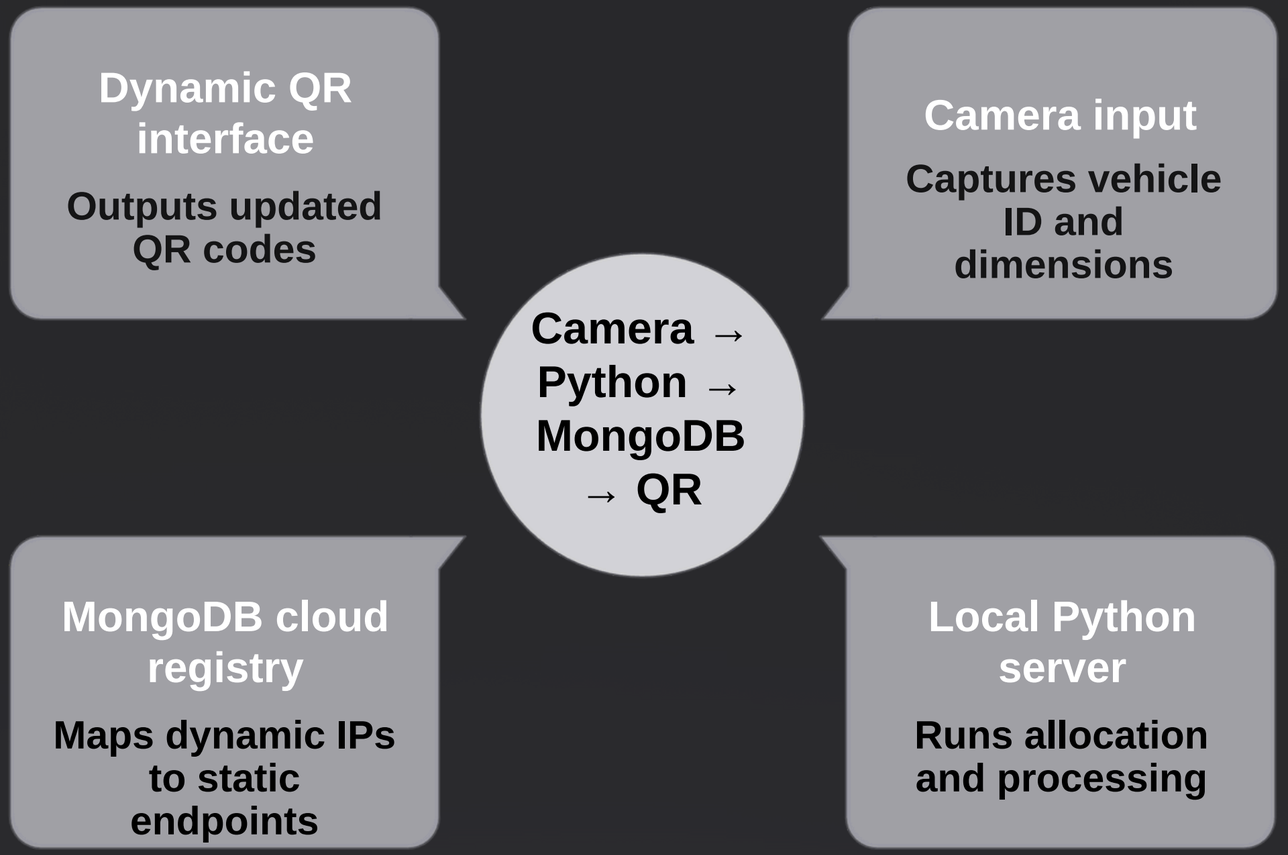
Connectivity

Cloud Registry ensures access across changing networks.

Differentiation:

- 95% Hardware Reduction: Replaces hundreds of sensors with one camera.
- Software-First: Infinite scalability without new wiring.

System Architecture: How it Works



Workflow:

1. **Input:** Camera captures vehicle identity and dimensions.
2. **Processing:** Local Python server runs allocation algorithms.
3. **Cloud Registry:** MongoDB maps dynamic local IPs to static endpoints.
4. **Interface:** Dynamic QR code redirects users to their assigned slot.

Feasibility: A Practical Approach

Practicality:

High. Uses standard Webcams, Laptops, and Free-tier Cloud (MongoDB).

Cost & Scalability:

- **Scalable:** Adding slots is a code update, not construction.
- **Economical:** Drastically lower installation costs vs. traditional IoT.

Challenges:

- **Issue:** Dynamic Network IPs.
- **Fix:** Automated Service Discovery via Cloud Registry.

Applications & Impact

Use Cases: Multi-level car parks, automated tolls, gated communities.

Developers

Lower infrastructure overhead.

Drivers

Zero search time; guaranteed parking.

Impact:

- **Tech:** Shifts industry from hardware-heavy to software-defined IoT.
- **Green:** Direct reduction in vehicle idling emissions.



Future Scope: Expanding Possibilities

Improvements:

- **Dynamic Pricing:** Demand-based automated rate adjustment.
- **Find My Car - Reverse Routing:** Using the system's entry logs to guide forgetful drivers back to their parked vehicle's exact location via the web interface.
- **Smart Energy Orchestration:** The system utilizes real-time occupancy data to automatically dim lights and deactivate ventilation in empty zones

Advanced Features:

- **Predictive Occupancy Modelling:** Our system logs entry timestamps to build a Predictive Model.
- **The "Blacklist" Security Protocol (ANPR):** Integrates with law enforcement databases to identify stolen vehicles, instantly triggering silent alarms and exit lockdowns.
- **Touchless Payment Integration:** RFID tag or Plate, calculates the fee, and auto-debits the user's wallet upon exit.

Optislot: Redefining Smart Parking for a Sustainable Future

Leveraging a revolutionary software-first approach, Optislot slashes hardware needs by 95%, delivering unmatched cost efficiency, effortless scalability, and profound environmental benefits. Gain a decisive competitive edge with a solution designed for tomorrow.

Partner with us to transform your parking infrastructure.

Reshekumar V - Project Lead and Developer

Email - vrk1711@gmail.com

LinkedIn - Reshekumar V

To know more about myself, give a view to my portfolio Reshekumar V